

AmplifiedTeacup: Trajectory Design in the VR Teacup Experience using HMD and Electric Wheelchair

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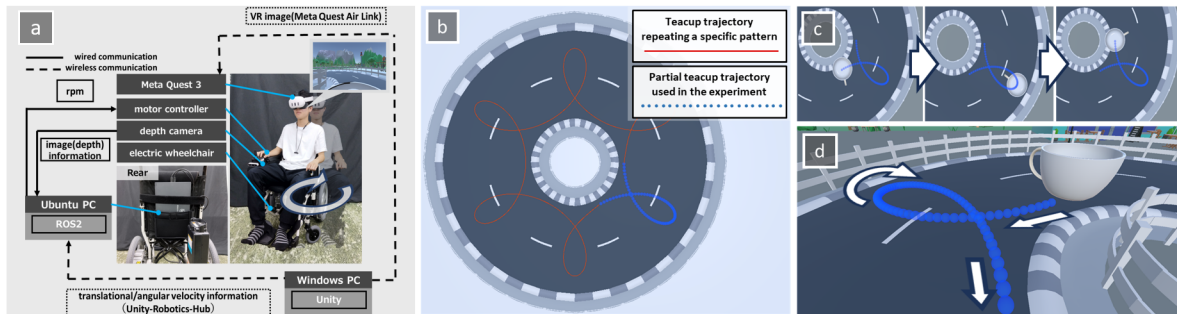


Figure 1: a) System diagram of this study, b) Teacup ride's trajectory used in this experiment, c) Temporal changes in the motion of the cup following the teacup attraction's trajectory as viewed from above, d) The cup moves along the trajectory of the teacup attraction in the virtual space.

ABSTRACT

Wheeled motion platforms (MPs) like small electric cars or electric wheelchairs have been combined with VR visuals via Head-Mounted Displays (HMDs) to create immersive experiences. However, space constraints and safety issues make it difficult to map virtual trajectories and speeds directly to physical movements. In this study, we investigate trajectories for a VR ride system combining HMDs and electric wheelchairs, focusing on the classic teacup amusement park ride. We evaluated multiple wheelchair trajectories under the same VR teacup visuals by systematically varying translational and rotational motion. Our findings suggest that wheelchair trajectories incorporating rotational speeds from the VR visuals have the potential to maintain user experience consistency, even with reduced translational speeds. This approach allows the wheelchair to rotate around a fixed point, which could be safer and less costly to operate, while still providing an immersive VR teacup experience.

Index Terms: Virtual Reality, Redirected Walking, Motion Platform

1 INTRODUCTION

Various attempts have been made to provide immersive VR rides by combining MPs with VR experience displayed on HMDs[4, 2]. Small electric wheelchairs, as MPs for VR rides, offer the advantage of moving riders with variable speed and acceleration, and their compact size allows for versatile use[1, 3]. However, space and safety constraints impose strict limitations on how virtual motion can be translated into physical trajectories for wheeled motion platforms such as electric wheelchairs.

In this study, we focused on the classic teacup ride and investigated the trajectories of an electric wheelchair as an MP for a VR ride system. The teacup ride generates complex trajectories through

the rotation of three discs (large, medium, small). By keeping the rotation radius, direction, and speed of each disc constant, repetitive trajectories are created (Figure 1 b). We developed a VR experience depicting such a repetitive trajectory and implemented nine distinct wheelchair trajectories by systematically varying the translational and angular velocities.

Our goal was to identify wheelchair trajectories that are smaller and safer than the reference trajectory in the virtual environment, while maintaining the quality of the VR teacup ride experience. A user study was conducted to evaluate each wheelchair trajectory under the same VR experience. Through this study, we investigated how much translational and angular motion could be reduced while preserving the quality of the VR teacup ride experience in a limited physical space.

2 SYSTEM DESIGN

2.1 Overview

The VR ride used in this study consists of an electric wheelchair (Yamaha Motor TOWNYJOY X PLUS+, drive wheel diameter 16 inches, caster wheel diameter 6 inches) and an HMD (Meta Quest 3) (Figure 1 a). The motor control is managed by a motor control device provided by Yamaha Motor for academic research and development purposes. The motor's rotational speed (rpm) is transmitted and received via serial communication with an Ubuntu PC (Ubuntu 22.04 LTS, GALLERIA GCR1660TGF-QCG) installed on the electric wheelchair. A depth camera (Intel RealSense D435i) is mounted to acquire visual odometry used for PID control.

2.1.1 Software Configuration

The HMD displays the VR visuals of the virtual teacup ride experience through real-time wireless communication (Meta Quest Air Link) from a Windows PC (Windows 11, GALLERIA GCR1660TGF-QCG). We used Unity for the visuals and ROS 2 (Robot Operating System 2) to send velocity information to the electric wheelchair, and Unity-Robotics-Hub integrates ROS 2 with Unity. The electric wheelchair is controlled via PID control integrating the translational and rotational velocity sent through ROS 2 and visual odometry obtained from the depth camera.

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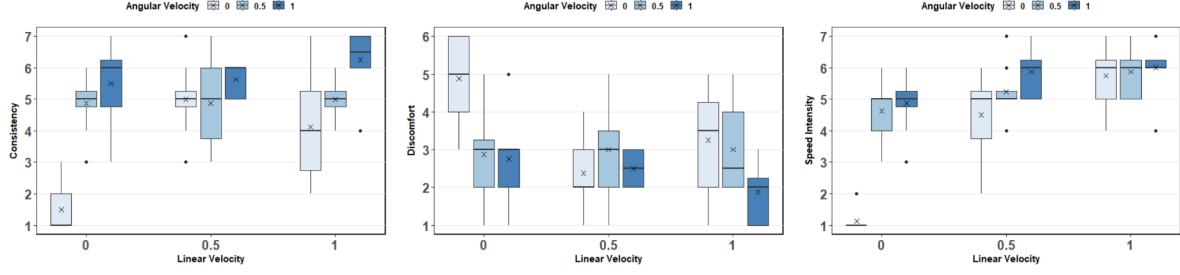


Figure 2: Subjective results on perceived visual-physical consistency, discomfort, and velocity intensity (from left to right).

2.1.2 Teacup Ride Trajectory

Teacup rides in amusement parks consist of large, medium, and small discs (cups). By adjusting their rotation radii, directions, and speeds, various cycloidal curves are produced. The position of a rider on the teacup ride can be expressed as:

$$\begin{aligned} x(t) &= R_1 \cos(\omega_1 t) + R_2 \cos(\omega_2 t) \\ y(t) &= R_1 \sin(\omega_1 t) + R_2 \sin(\omega_2 t) \end{aligned}$$

where R_1 is the rotation radius of the medium disc, R_2 is the distance from the center of the medium disc to the rider's position, ω_1 and ω_2 are the angular velocities of the large and medium discs, respectively, and t is the elapsed time. The small disc is fixed to the medium disc, creating a repeated cycloidal trajectory.

2.1.3 VR Teacup Visual Experience

The VR video of the teacup ride was created using Unity 2021.3.26f1. The variables were set to $R_1 = 2.5$ m, $R_2 = 1.0$ m, $\omega_1 = 0.0698$ rad/s, and $\omega_2 = 0.349$ rad/s, based on actual amusement park teacup rides. The discs' radii were set to 4 m, 1.5 m, and 0.5 m, with rotations set to counterclockwise, clockwise, and no rotation, respectively. The visual content presented on the HMD depicts movement along the cycloidal trajectory used in this study.

3 USER STUDY

We conducted a user study with 8 participants (5 males and 3 females) to investigate how much physical space is needed for a power wheelchair to move around. Participants experienced the same VR teacup ride and compared its visual motion with wheelchair trajectories moving at the same or reduced speeds.

In this study, rather than identifying perceptual detection thresholds in redirected walking, we focus on practical design decisions for reducing physical motion in a wheelchair-based VR ride, using the teacup attraction as a case involving non-steady translational and rotational motion with acceleration. Based on this characteristic, the physical motion of the electric wheelchair was defined as a scaling of the VR visual motion to gain insight into the amount of physical motion required for such trajectories. The translational and angular velocities of the electric wheelchair (v and ω) were controlled as scaling factors of the corresponding velocities in the VR visual motion to broadly assess the amount of physical motion required. Specifically, scaling factors of 1.0 (matching the VR velocity), 0.5 (half of the VR velocity), and 0.0 (no physical motion) were applied, resulting in nine experimental conditions (3×3).

After each trial, participants rated the consistency between the VR visual motion and the wheelchair trajectory, their perceived discomfort during the VR experience, and the perceived velocity intensity of the electric wheelchair using a seven-point Likert scale (1 = very low, 7 = very high). For discomfort, higher scores indicated greater discomfort, reflecting a stronger perceived mismatch between the VR visual motion and the physical trajectory.

Velocity type	Pair	Consistency p-value	Discomfort p-value	Velocity intensity p-value
$\omega \times 0.0$	$v \times 0.0$ vs $v \times 0.5$	.008 **	.008 **	.008 **
	$v \times 0.0$ vs $v \times 1.0$	.016 *	.063 †	.008 **
	$v \times 0.5$ vs $v \times 1.0$	.250	.250	.125
$\omega \times 0.5$	$v \times 0.0$ vs $v \times 0.5$	1.000	1.000	.250
	$v \times 0.0$ vs $v \times 1.0$	1.000	1.000	.125
	$v \times 0.5$ vs $v \times 1.0$	.984	1.000	.250
$\omega \times 1.0$	$v \times 0.0$ vs $v \times 0.5$	1.000	.750	1.000
	$v \times 0.0$ vs $v \times 1.0$	.406	.125	.406
	$v \times 0.5$ vs $v \times 1.0$	.281	.375	.281
$v \times 0.0$	$\omega \times 0.0$ vs $\omega \times 0.5$	.008 **	.016 *	.008 **
	$\omega \times 0.0$ vs $\omega \times 1.0$	.008 **	.004 †	.063 †
	$\omega \times 0.5$ vs $\omega \times 1.0$	.313	.250	.125
$v \times 0.5$	$\omega \times 0.0$ vs $\omega \times 0.5$	1.000	1.000	.250
	$\omega \times 0.0$ vs $\omega \times 1.0$	.313	1.000	.438
	$\omega \times 0.5$ vs $\omega \times 1.0$	.313	1.000	.313
$v \times 1.0$	$\omega \times 0.0$ vs $\omega \times 0.5$	.266	.125	.750
	$\omega \times 0.0$ vs $\omega \times 1.0$	.031 *	.375	1.000
	$\omega \times 0.5$ vs $\omega \times 1.0$	.016 *	.750	1.000

Figure 3: Pairwise comparison results showing significant differences in perceived consistency, velocity intensity, and discomfort. * $p < 0.05$, ** $p < 0.01$, † $p < 0.1$.

Analysis of variance revealed significant main effects and interactions, followed by post-hoc tests. The results showed that reducing angular velocity (ω) while keeping translational velocity (v) at 1.0 led to significant differences in consistency, whereas reducing v while keeping ω at 1.0 did not. These results indicate that angular velocity plays a more critical role than translational velocity in maintaining consistency between the physical trajectory and the VR visual motion.

4 CONCLUSION

Taken together, these findings suggest that a trajectory in which the electric wheelchair rotates around a fixed point could safely provide a highly consistent VR teacup experience, even in a small physical space. While the present study employed a simplified teacup model, future work may incorporate additional rotational components corresponding to the rotation of the small disc, such as continuous yaw rotation of the rider's viewpoint along the trajectory, to achieve a more realistic and immersive experience. This extension may further enhance the potential of redirection-based VR ride design.

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